



**DAM**

# **WEB INTERFACE DESIGN**

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# BOOK



## Diseño de interfaces web

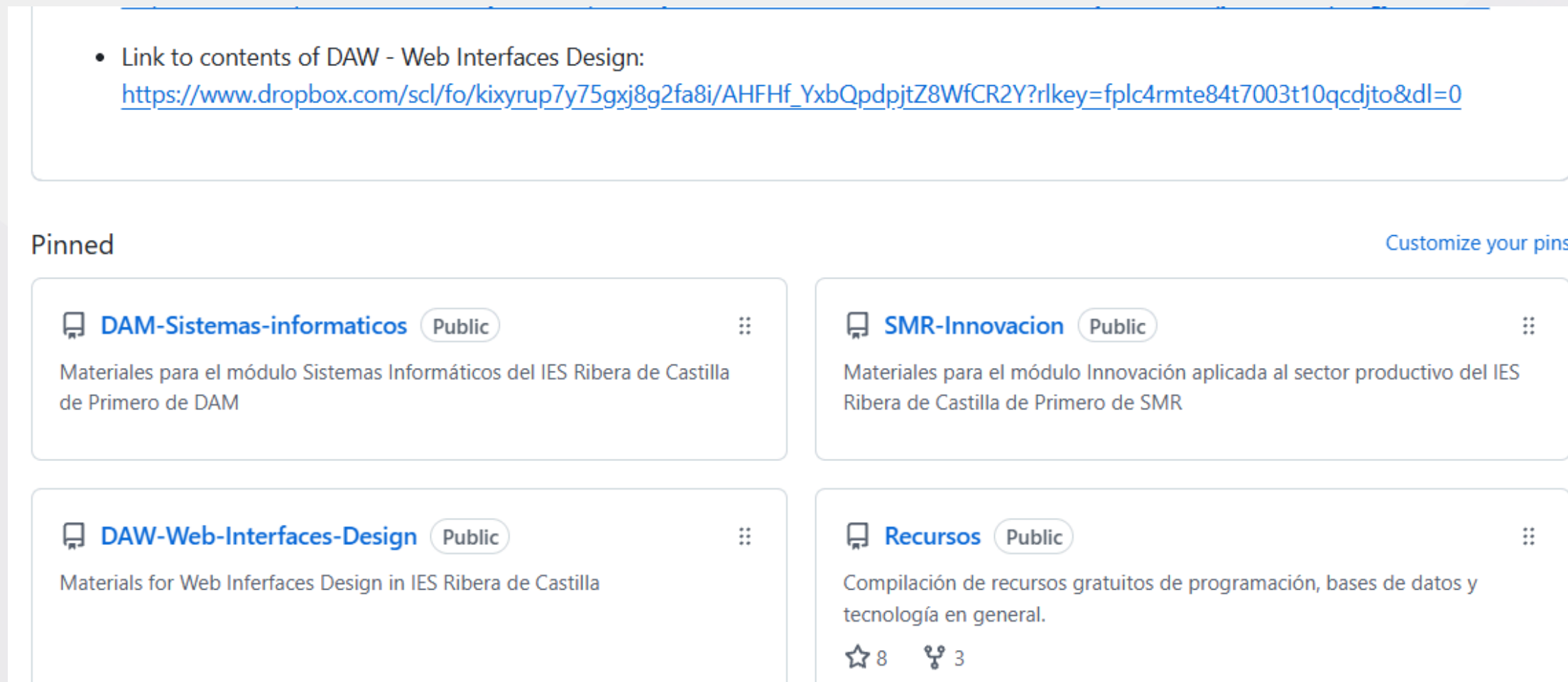
2.ª edición  
revisada y actualizada

# MATERIALS

GitHub: <https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design>

Dropbox:

[https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf\\_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0](https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0)



The screenshot shows a user's profile page with a section titled "Pinned". At the top of this section, there is a link to the contents of DAW - Web Interfaces Design: [https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf\\_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0](https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0). Below this, there are four pinned items arranged in a 2x2 grid. Each item has a folder icon, a title, a "Public" status, and a description. The items are: 1. "DAM-Sistemas-informaticos" (Public) with description "Materiales para el módulo Sistemas Informáticos del IES Ribera de Castilla de Primero de DAM". 2. "SMR-Innovacion" (Public) with description "Materiales para el módulo Innovación aplicada al sector productivo del IES Ribera de Castilla de Primero de SMR". 3. "DAW-Web-Interfaces-Design" (Public) with description "Materials for Web Interfaces Design in IES Ribera de Castilla". 4. "Recursos" (Public) with description "Compilación de recursos gratuitos de programación, bases de datos y tecnología en general." and engagement metrics of 8 stars and 3 shares.

- Link to contents of DAW - Web Interfaces Design:  
[https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf\\_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0](https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0)

Pinned [Customize your pins](#)

 **DAM-Sistemas-informaticos** Public  
Materiales para el módulo Sistemas Informáticos del IES Ribera de Castilla de Primero de DAM

 **SMR-Innovacion** Public  
Materiales para el módulo Innovación aplicada al sector productivo del IES Ribera de Castilla de Primero de SMR

 **DAW-Web-Interfaces-Design** Public  
Materials for Web Interfaces Design in IES Ribera de Castilla

 **Recursos** Public  
Compilación de recursos gratuitos de programación, bases de datos y tecnología en general.  
★ 8    🔗 3

# TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:

|4s6zs6

# USER INTERFACE DESIGN FOR WEB DEVELOPMENT

## Goals:

- Recognize the importance of visual communication and its basic principles.
- Analyze different alternatives for presenting information in web documents.
- Identify and analyze the elements involved in creating web prototypes.
- Use and analyze content management systems as the foundation for developing a website.
- Plan and establish the process for designing a website.

# DESIGN PRINCIPLES BASED ON HUMAN-COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a field of study that focuses on the design and development of interactive systems that are easy and efficient to use.

HCI guidelines are recommendations that can be used to design and develop user interfaces that are more user-friendly and effective.

# LICKLIDER & CLARK (1962)

Ten **problems to solve** for better Human-Computer Interaction (HCI):

- Sharing computer time among many users.
- An input-output system for communication using symbolic and graphical data.
- An interactive system capable of processing operations in real time.
- Systems for massive information storage that allow rapid retrieval.
- Systems that facilitate cooperation among people in the design and programming of large systems.

# LICKLIDER & CLARK (1962) (II)

Ten **problems to solve** for better Human–Computer Interaction (HCI) (II):

- Recognition by computers of voice, printed handwriting, and direct handwritten data input.
- Automatic translation between natural languages.
- Automatic programming — enabling computers to generate or assist in writing programs.
- Self-organizing systems capable of adapting and improving their own performance.
- Logical and heuristic problem-solving systems that support human reasoning.

# LICKLIDER & CLARK (1962) (III)

These ten problems anticipated almost everything modern computing tries to solve:

- multitasking
- graphical interfaces
- real-time systems
- databases
- collaborative development
- speech recognition
- machine translation
- AI-assisted coding
- adaptive systems
- expert systems

# HANSEN (1972)

Wilfred J. Hansen's four principles for user-interface design, proposed in 1971, they are an important early contribution to Human-Computer Interaction (HCI).

Hansen presented them in “User Engineering Principles for Interactive Systems”, published in the proceedings of the 1971 Fall Joint Computer Conference.

The four principles are:

1. Know the user
2. Minimize memorization
3. Optimize operations
4. Engineer for errors

# 1. KNOW THE USER

The interface should be designed around the people who will actually use it, rather than around what is technically convenient for the programmer.

The designer should understand:

- the user's goals and needs;
- their previous experience;
- their knowledge and skills;
- the context in which they work;
- their limitations and likely difficulties.

## 2. MINIMIZE MEMORIZATION

Users should not have to remember unnecessary information in order to operate the system.

The interface should provide information that the user needs at the moment they need it.

For example, instead of requiring:

*"Remember that Ctrl+Alt+N creates a new network."*

the system could provide a visible New network button or menu item.

This principle is closely related to a major modern HCI idea: recognition is generally easier than recall.

## 2. MINIMIZE MEMORIZATION (II)

You can see its influence in things we now take for granted:

- menus;
- visible buttons;
- autocomplete;
- contextual help;
- suggestions;
- recently used items;
- meaningful icons and labels.

# 3. OPTIMIZE OPERATIONS

The system should make users' actual tasks efficient.

This means reducing unnecessary actions, avoiding cumbersome procedures and designing operations around the user's workflow.

Design the system so that the user's work can be accomplished efficiently and naturally.

This is particularly relevant today for usability, keyboard shortcuts, automation, defaults and reducing the number of steps required to accomplish frequent tasks.

## 4. ENGINEER FOR ERRORS

Hansen recognized that users will make mistakes, and that a good system should therefore be designed to anticipate and handle them.

The important point is that errors should not simply be treated as the user's fault. For example, if a user enters: 192.168.1.300  
a good interface should detect the invalid address and explain the problem.

An even better interface might prevent invalid input from being entered in the first place.

# 4. ENGINEER FOR ERRORS (II)

This principle leads to several modern practices:

- validation.
- meaningful error messages.
- confirmation for dangerous operations.
- undo.
- recovery mechanisms.
- preventing invalid actions.
- preserving user data after an error.

# CORE FUNCTIONS OF GRAPHIC DESIGN

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **Communication & Information Delivery:** Translates raw data, copy, or abstract concepts into structured, easily digestible visual messages (think infographics, road signs, or annual reports).
- **Persuasion & Marketing:** Drives action and influences audience behavior in advertising, campaigns, and product packaging by appealing to specific emotions and needs.
- **Identity & Brand Building:** Establishes the distinct visual personality of a company, product, or organization through consistent logos, typography, color schemes, and imagery.

# CORE FUNCTIONS OF GRAPHIC DESIGN (II)

Graphic design isn't just "making things pretty." In web development, it has functional, communicative, and cognitive purposes.

- **User Experience (UX) & Navigation:** Guides people through digital spaces (websites, apps) and physical environments (wayfinding in airports or subways) by creating intuitive visual hierarchies.
- **Engagement & Aesthetics:** Captures attention in a visual-heavy world, making content memorable and pleasant to interact with.

# GROUPS OF ELEMENTS

- **Conceptual elements:** The ideas, concepts and meanings that the interface represents. Example: the concept of "deleting a file."
- **Visual elements:** What the user actually perceives: colours, typography, icons, shapes, images, etc. Example: a trash-bin icon representing deletion.
- **Relational elements:** The relationships between elements: hierarchy, proximity, alignment, grouping, sequence and navigation. Example: placing "Save" next to "Cancel", or grouping network settings together.
- **Practical elements:** The elements that allow the user to perform actions and accomplish tasks. Example: buttons, forms, sliders, menus and controls.

# DEFINITION OF WEB INTERFACE

For all the above, we can conclude that:

*“A web interface is the set of **visual and interactive elements**, together with their **organization and layout**, that allows users to access information, navigate a website and interact with its content and functions efficiently.”*

If these elements are not carefully designed and coordinated, a website may still work technically, but achieving a good user experience will largely be a matter of chance.

# EXAMPLES OF GOOD AND BAD DESIGN

**DE JUEVES A SABADOS DE 20.30 A 00.00 HRS.** **SABADOS Y**  
**DOMINGOS DE 12.30 A 14.30 HRS.**

**FUSILES AL FERRETTO**

**RABAS A LA PROVENZAL**

**CARACOLLES A LA BORDALEZA**

**RABAS A LA CALABRIA**

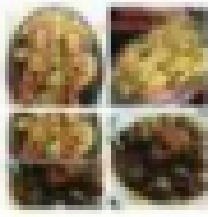
**BERLUZA AL AJILLO**

**GAMBAS AL AJILLO**

**ABIERTO DE JUEVES A SABADOS DE NOCHE- SABADOS Y DOMINGOS ALMUERZOS**

**HORARIO CENA DE 20.30 A 23.30 HRS.**

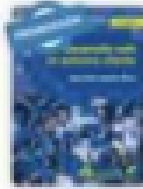
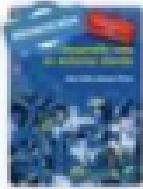
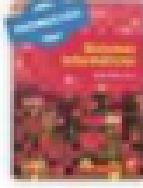
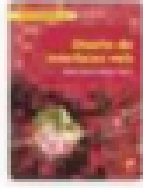
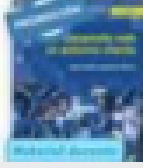
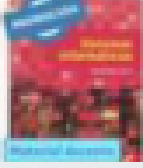
**HORARIO ALMUERZO DE 12.30 A 14.30 HRS.**



**EDITORIAL SINTESIS**

ENTRAR POR TITULO, AUTOR O ISBN

ORDENAR POR: **ULTIMOS LIBROS**

|                                                                                       |                                                                                                                                              |                                                                                       |                                                                                                                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|    | Monica Pérez, Juan Carlos<br>9788410177008<br>Hasta 01/01/24<br><b>Desarrollo web en entorno cliente (2.ª edición)</b><br>142 páginas de 204 |    | Monica Pérez, Juan Carlos<br>9788410177008<br>Hasta 01/01/24<br><b>Desarrollo web en entorno cliente (2.ª edición)</b><br>142 páginas de 204 |
|   | Rosa Arce, Jesús<br>9788410177015<br>Hasta 01/01/24<br><b>Sistemas Informáticos</b><br>42 páginas de 106                                     |   | Gerardo Miguel López, Diana<br>9788410177015<br>Hasta 01/01/24<br><b>Diseño de interfaces web</b><br>11 páginas de 20                        |
|  | Monica Pérez, Juan Carlos<br>9788410177008<br>Hasta 01/01/24<br><b>Desarrollo web en entorno cliente (2.ª edición)</b><br>142 páginas de 204 |  | Rosa Arce, Jesús<br>9788410177015<br>Hasta 01/01/24<br><b>Sistemas Informáticos</b><br>42 páginas de 106                                     |

# THE 6 PHASES OF WEBSITE DESIGN

The web design process moves from abstract strategy to concrete implementation in six distinct stages.

## 01. STRATEGY

Discovery, goals, & audience definition

## 02. STRUCTURE

Sitemap, user flows, & wireframes

## 03. UI DESIGN

Visual style, design system, & mockups

## 04. BUILD

Front-end, back-end, & CMS integration

## 05. TEST

QA, performance, accessibility, & cross-browser

## 06. LAUNCH

Deployment & continuous iteration

# PHASE 1: DISCOVERY & STRATEGY

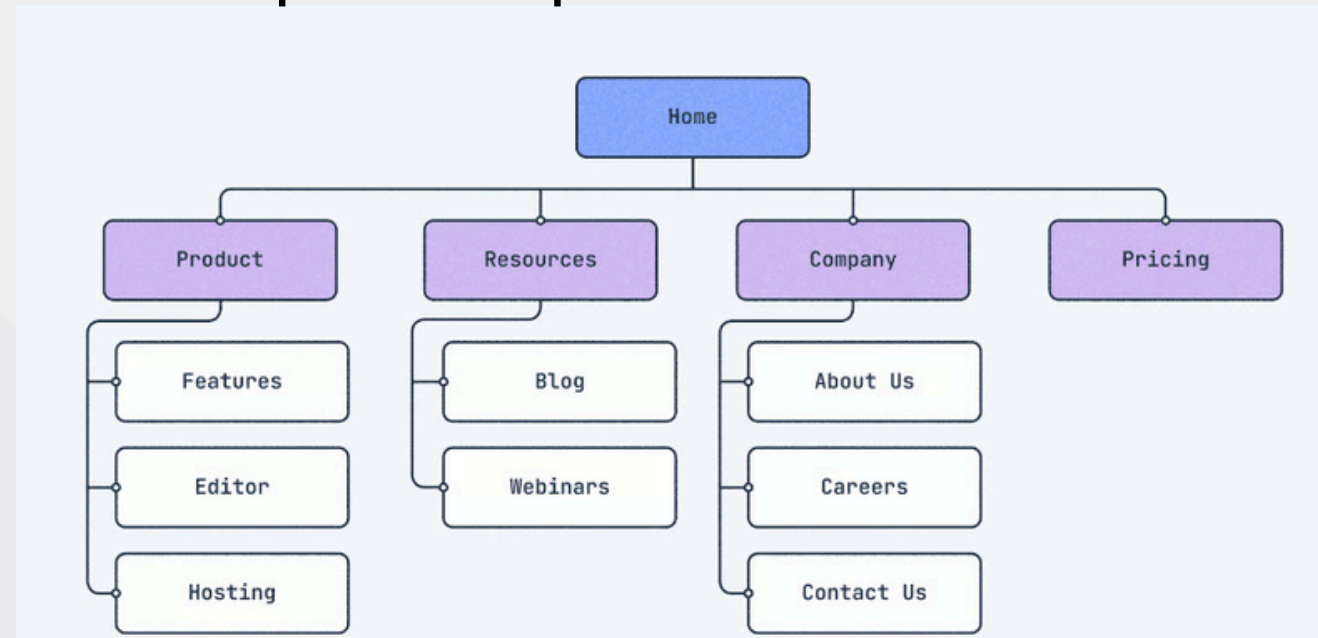
- **Objective:** Define the project scope, business goals, and target audience.
- **Key Tasks:**
  - Conduct stakeholder interviews
  - **Analyze competitors**
  - **Create user personas**
  - Establish Key Performance Indicators (KPIs).

# PHASE 2: ARCHITECTURE & WIREFRAMING (UX)

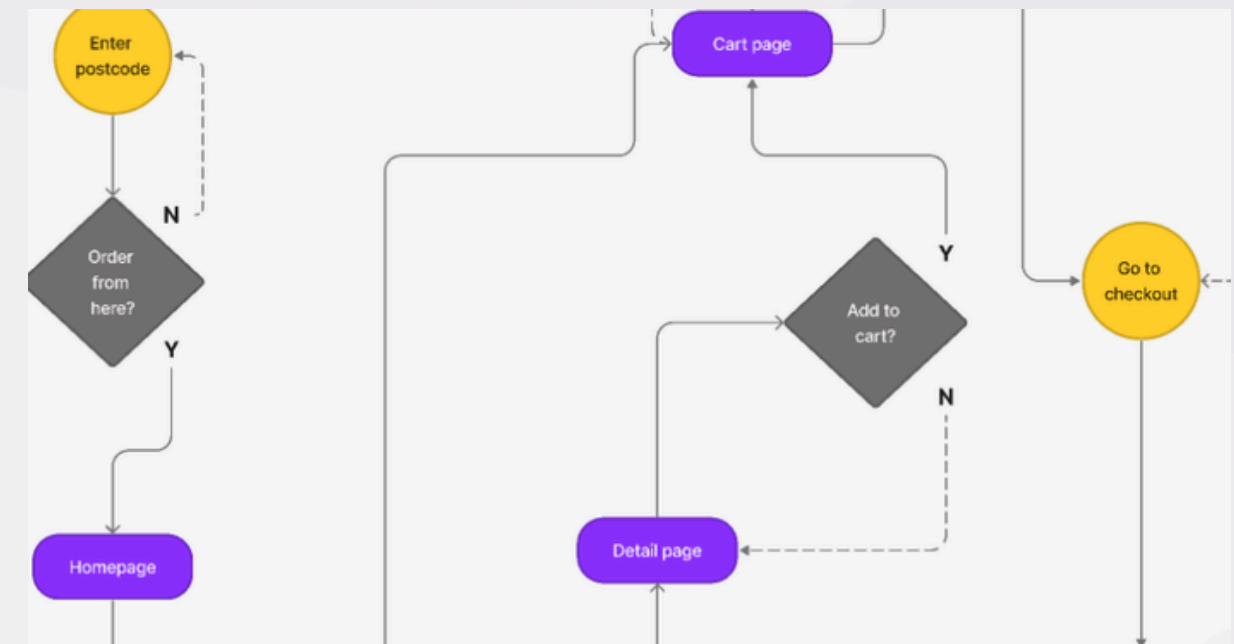
- **Objective:** Map out how content is organized and how users navigate the site.
- **Key Deliverables:**
  - **Sitemap:** A structural tree of all pages.
  - **User Flows:** Step-by-step paths users take to complete key tasks (e.g., checkout, signup).
  - **Wireframes:** Low-fidelity grayscale blueprints focusing on element placement without visual styling.

# PHASE 2: ARCHITECTURE & WIREFRAMING (UX) (II)

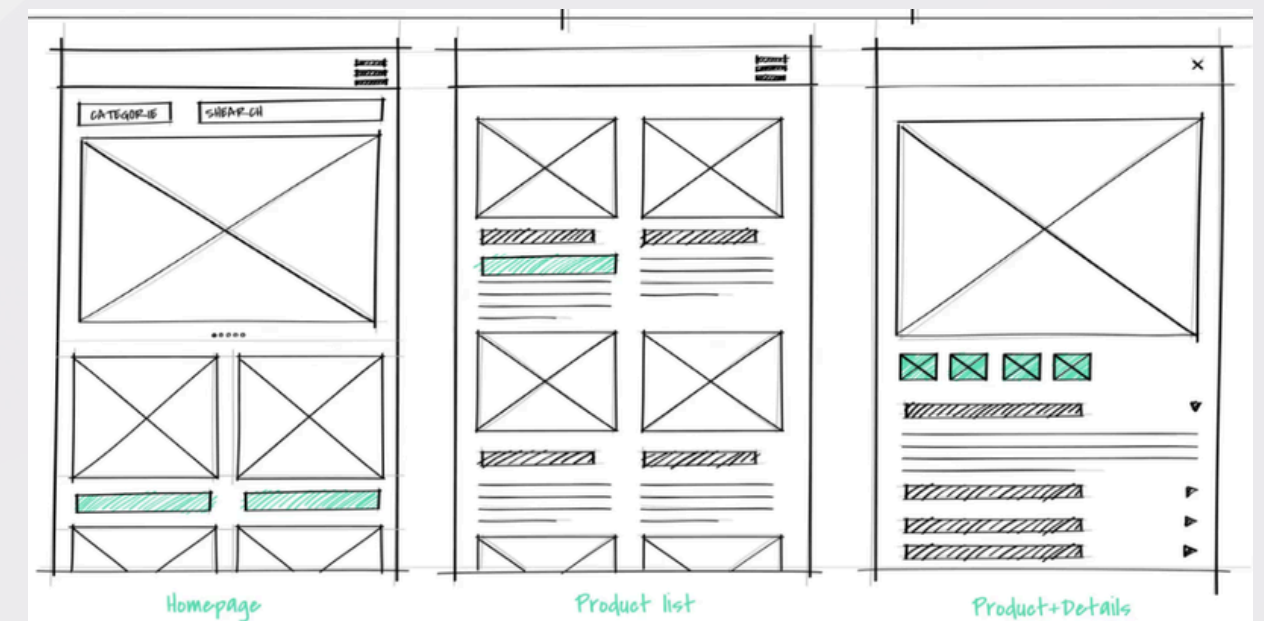
Sitemap example



User flow



Wireframe

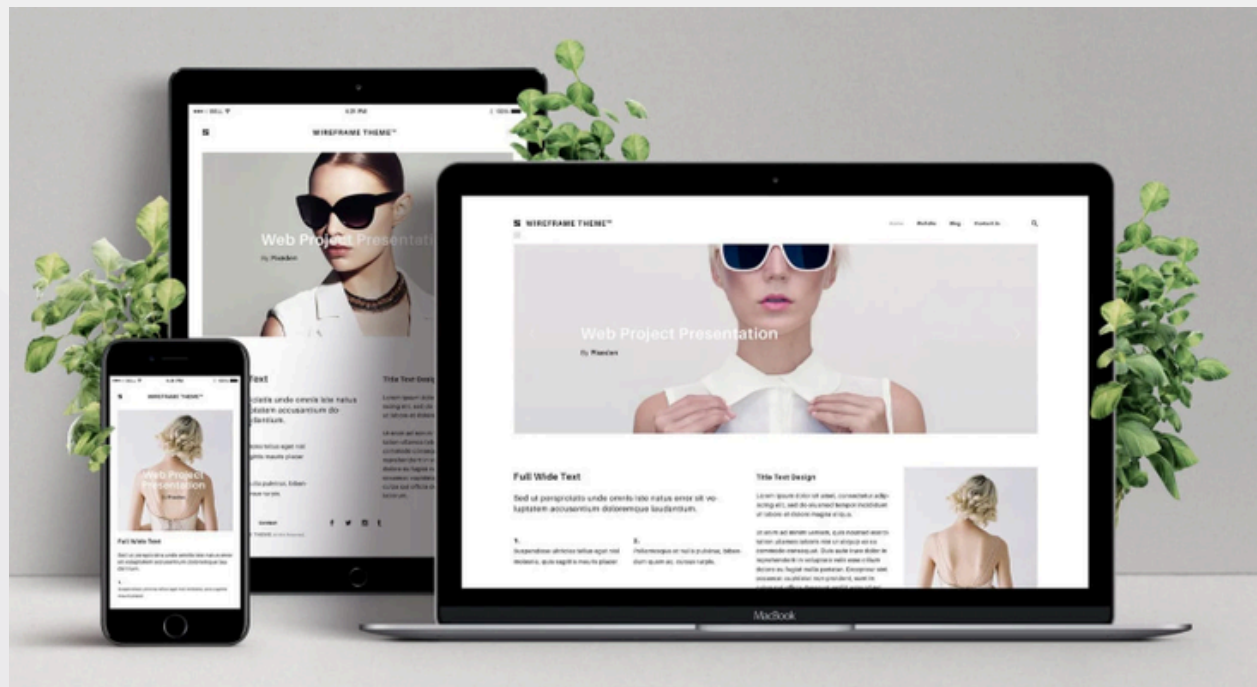


# PHASE 3: VISUAL DESIGN (UI)

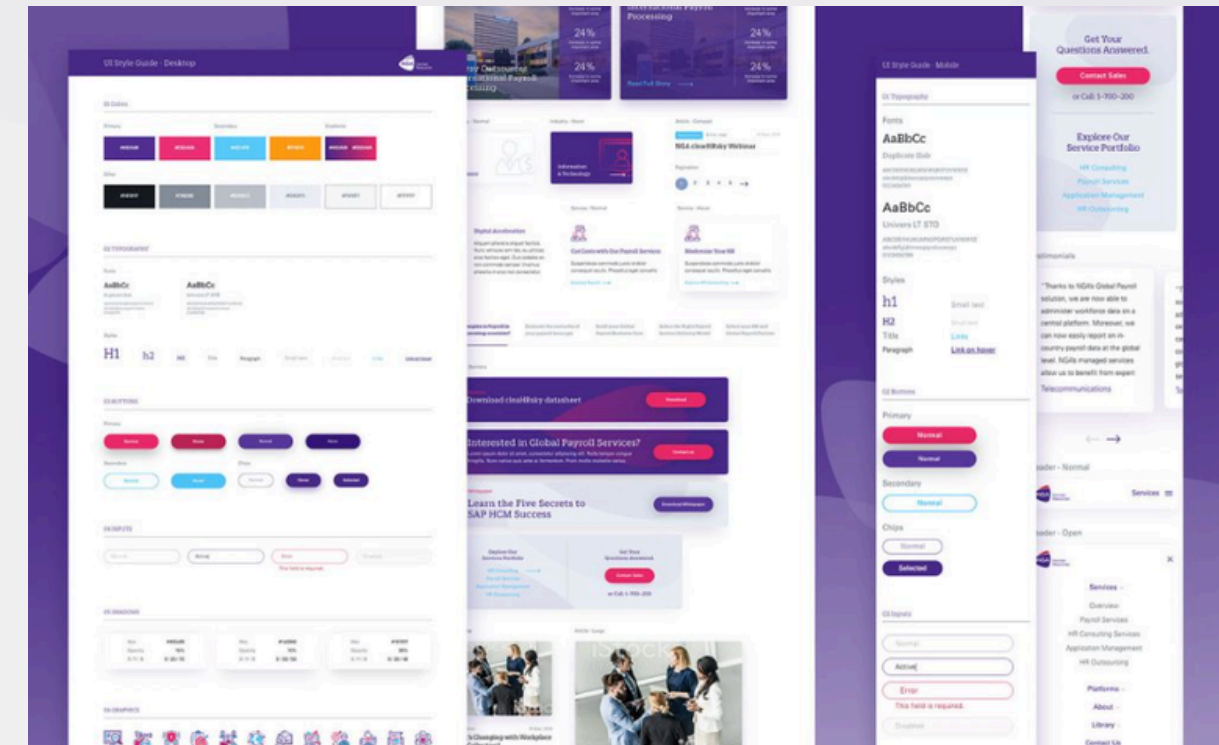
- **Objective:** Apply brand identity, color theory, and visual polish to wireframes.
- **Key Deliverables:**
  - High-fidelity design **mockups**.
  - **Responsive layouts** (desktop, tablet, mobile).
  - **Interactive prototypes**.
  - A **design system** or **component library**.

# PHASE 3: VISUAL DESIGN (UI) (II)

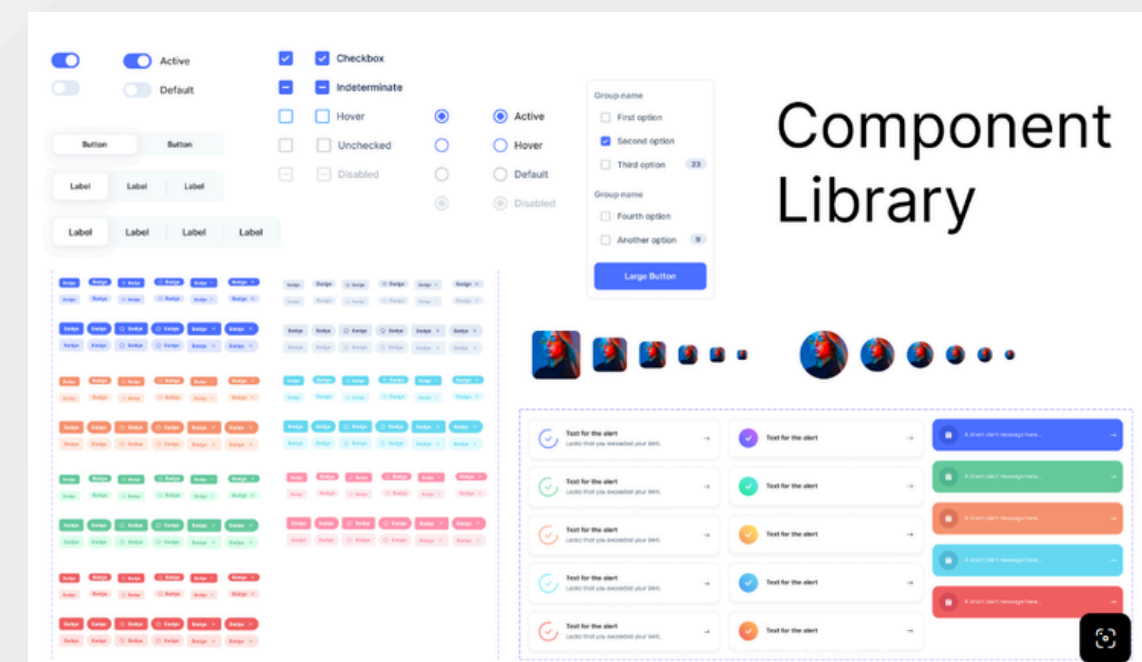
## Responsive Layouts



## Design system



## Component library



## Component Library

# PHASE 4: DEVELOPMENT

- **Objective:** Turn visual designs into functional code.
- **Key Tasks:**
  - Front-end development (HTML, CSS, JavaScript/frameworks)
  - Back-end/API setup
  - Database architecture
  - CMS integration (e.g., WordPress, Webflow, custom headless setup).

# PHASE 5: TESTING & QUALITY ASSURANCE (QA)

- **Objective:** Ensure functional integrity, performance, and accessibility across all platforms.
- **Key Tasks:**
  - Cross-browser and cross-device responsiveness.
  - Web accessibility compliance (WCAG 2.1 AA).
  - Load time optimization and technical SEO best practices.
  - Form submissions and interactive feature validation.

# PHASE 6: LAUNCH & OPTIMIZATION

- **Objective:** Deploy the site to production and monitor performance.
- **Key Tasks:**
  - Domain migration
  - SSL setup, analytics tracking setup (e.g., Google Analytics)
  - post-launch bug fixes
  - continuous conversion rate optimization (CRO).

# TASK 1

Define the project scope, business goals, and target audience for your **Intermodular Project**.

- **Key Tasks:**
  - **Analyze competitors:** at least look for two webpages from direct competition in the area and identify the sitemap. We will analyze their design system later.
  - **Create user personas**
  - Establish Key Performance Indicators (KPIs): for example, client satisfaction.

# **APPLICATIONS: ANALYSIS, DESIGN, AND PROTOTYPES**

Why mockups and prototypes matter

- They define the structure and layout early
- They accelerate development
- They help clients visualize the final product
- They reduce misunderstandings and rework

# PROTOTYPES. KEY ELEMENTS

- A prototype is a mockup or model of a design that helps visualize the final product.
- Prototypes allow designers to explore different layout options, test basic functionality, and evaluate usability.
- They save time, effort, and money, since it is much easier to make changes to an early design than to modify the final product.

# WHY PROTOTYPES MATTER IN WEB DESIGN

- They allow multiple design iterations
- They help refine menus and interface elements
- They involve the client early in the process
- They enable functionality and usability testing
- They reduce cost and rework before coding begins

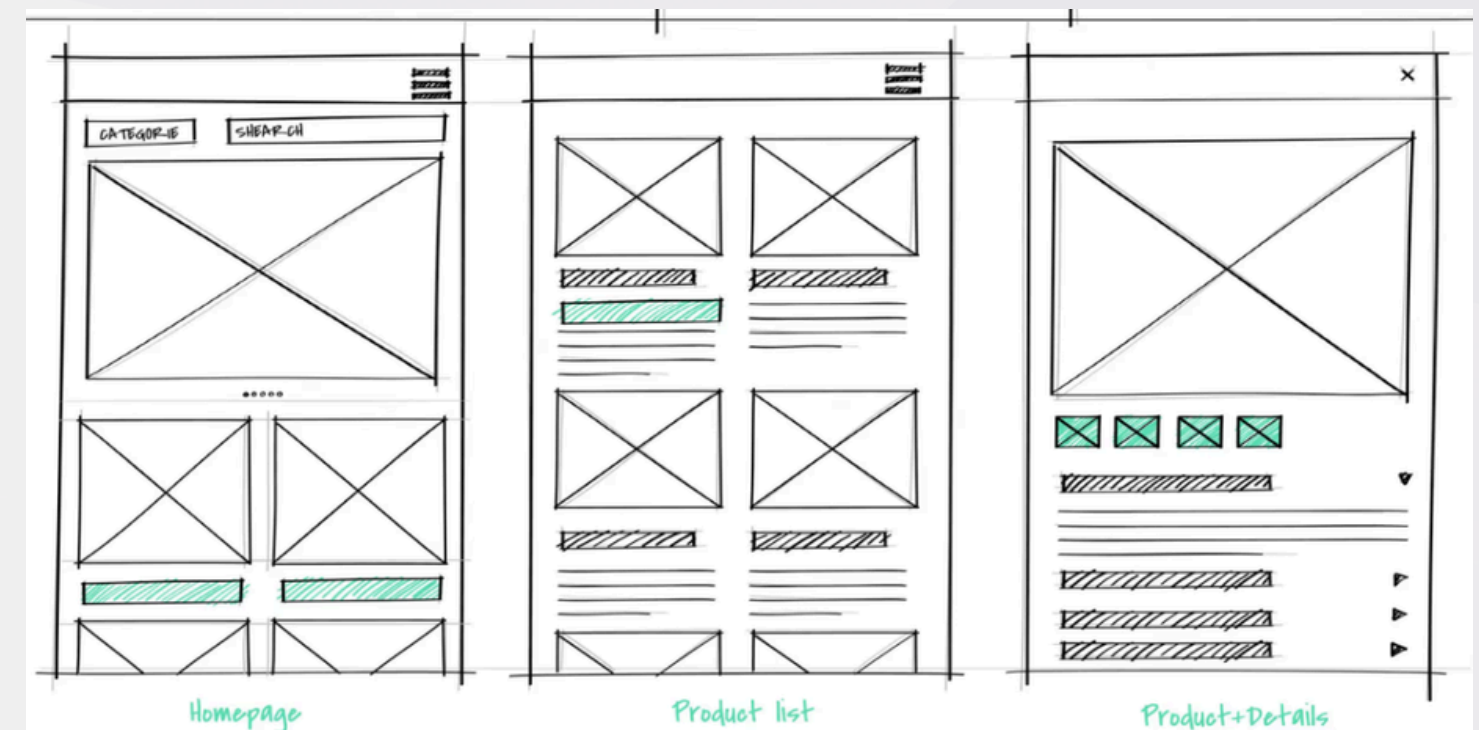
# TYPES OF PROTOTYPES: SKETCHING

- Paper-based drawings of the whole website
- Includes processes and screen relationships
- Used in the early design phase
- Defines content hierarchy
- No visual design details yet



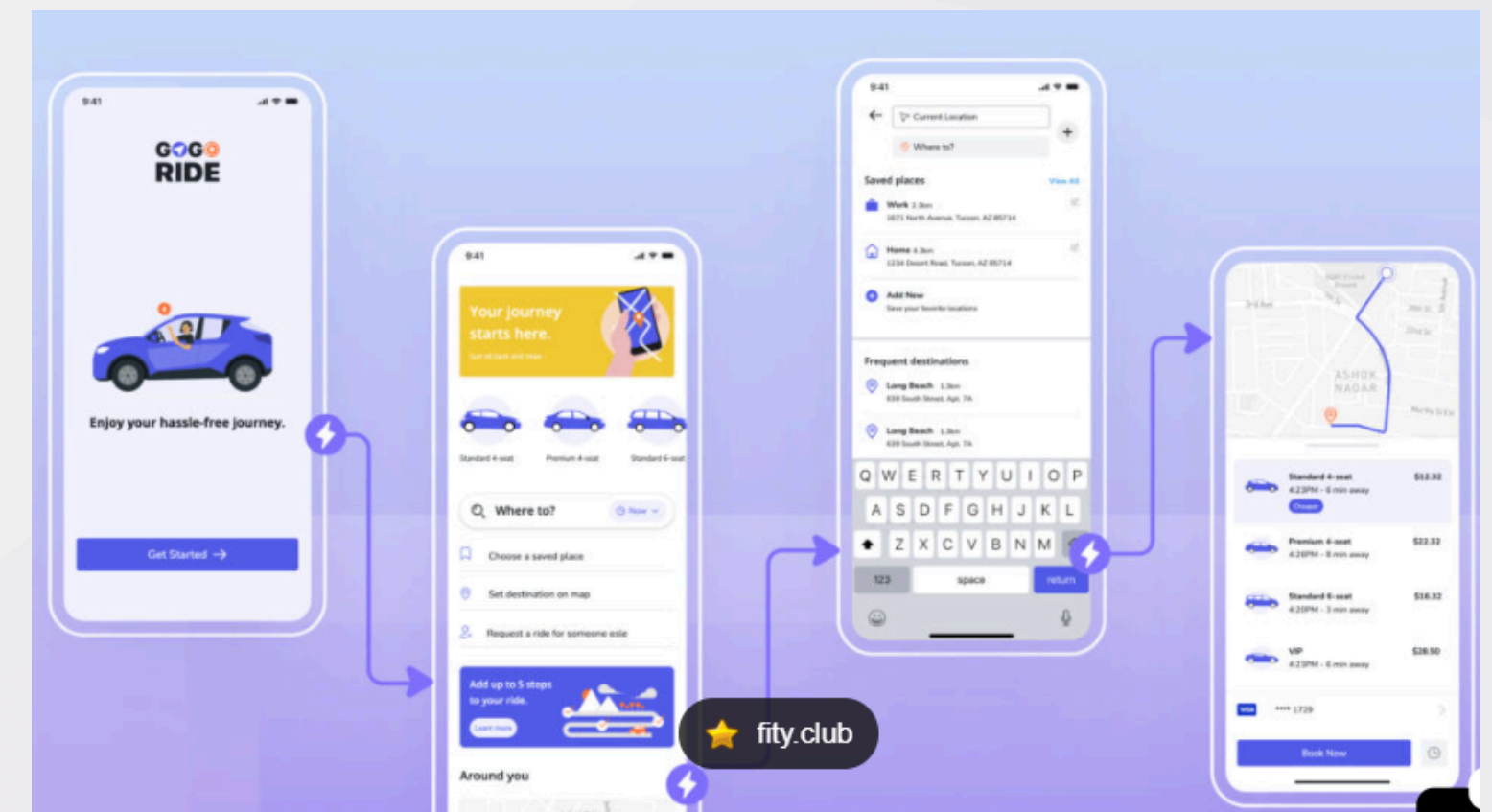
# TYPES OF PROTOTYPES: WIREFRAMING

- More detailed screen drawings
- Includes content structure and **calls to action**
- Paper or digital format
- Produces a mock-up based on earlier sketches
- Helps validate design decisions with the client
- Main goal: content organization



# TYPES OF PROTOTYPES: PROTOTYPING

- Digital design of screen interactions
- Final step before full development
- Evaluates design, content flow, and functionality
- Includes menus, forms, buttons, icons...
- Enables user testing
- Reduces development time and number of app versions



# WIREFRAMING ORGANIZATION

The design of the prototype, in any of its phases, should be based mainly on the following aspects:

- Identifying the elements that belong to each page of the website.
- Distributing the number of graphical interface elements so that the screen is not overloaded, while still providing enough information and ensuring correct interaction.
- Organizing the hierarchy, order, and arrangement of elements.
- Ensuring an appropriate layout that efficiently uses the available space depending on the device.
- Selecting web design patterns to standardize the interface design.
- Evaluating technical aspects of usability and accessibility.

# TASK 2A

Create a sketching and wireframing to design the user interface for the inter-module project (including mobile devices).

Use pencil (or pen) to quickly sketch the different screens and user flows, including the home screen.

Experiment with different layouts and design elements

# RESOURCES

Unplash (Royalty Free) images: <https://unsplash.com/es>

Google Fonts: <https://fonts.google.com/>

# ADDITIONAL READINGS

[Guidelines in Human Computer Interface \(HCI\)](#)

[Interaction in HCI: Principles, types, and examples](#)

[How to create wireframes as a UX Designer](#)

[Design Systems 101: What are they and why do they matter?](#)

[How to Create a Website Wireframe in Figma \(Beginner Tutorial\)](#)

[How to Wireframe in Figma in 2026](#)

# ACRONIMS

**HCI:** Human-Computer Interaction

**I/O or IO:** Input/Output

**SME:** Small and medium-sized enterprises

**UX:** User Experience

**UI:** User Interface

**¿ANY QUESTION?**